

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

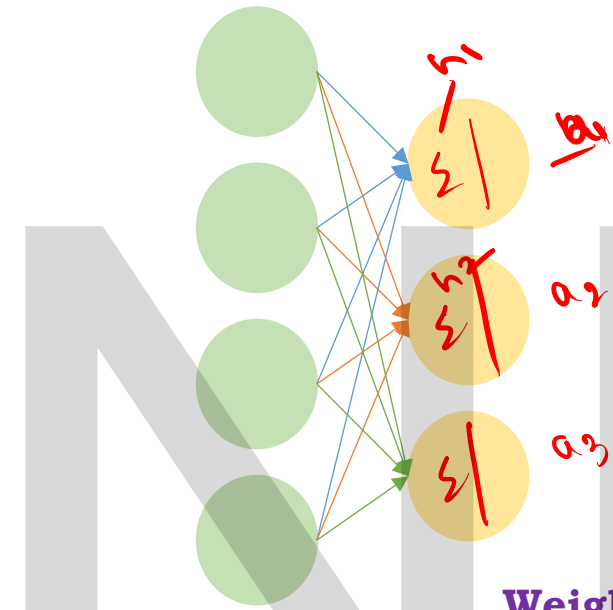
Week 3: Variational Autoencoders (VAEs): Foundations

Today's Session Overview

- Understand VAE working using a numerical example

NPTREL

Variational Autoencoders (VAEs): Encoder



Encoder Forward Pass

Input:
 $x = [5.1, 3.5, 1.4, 0.2]$

Let Bias (b_1) = 0.1 $h = xW_1 + \underline{b_1}$

Weights W_1 (4×3): $\begin{bmatrix} 0.1, -0.2, 0.3, \\ 0.4, 0.5, -0.6, \\ -0.7, 0.8, 0.9, \\ 0.2, -0.1, 0.4 \end{bmatrix}$

$\max(0, h)$
 $\max(0, 1.07)$
 $\max(0, 1.93)$
 $\max(0, 0.87)$

$$h_1 = (5.1 \times 0.1) + (3.5 \times 0.4) + (1.4 \times -0.7) + (0.2 \times 0.2) + 0.1 = \underline{1.07}$$

$$h_2 = (5.1 \times -0.2) + (3.5 \times 0.5) + (1.4 \times 0.8) + (0.2 \times -0.1) + 0.1 = \underline{1.93}$$

$$h_3 = (5.1 \times 0.3) + (3.5 \times -0.6) + (1.4 \times 0.9) + (0.2 \times 0.4) + 0.1 = \underline{0.87}$$

ReLU activation is applied to the hidden layer:

$$a_i = \max(0, h_i) \checkmark$$

Since all the computed hidden values are positive,

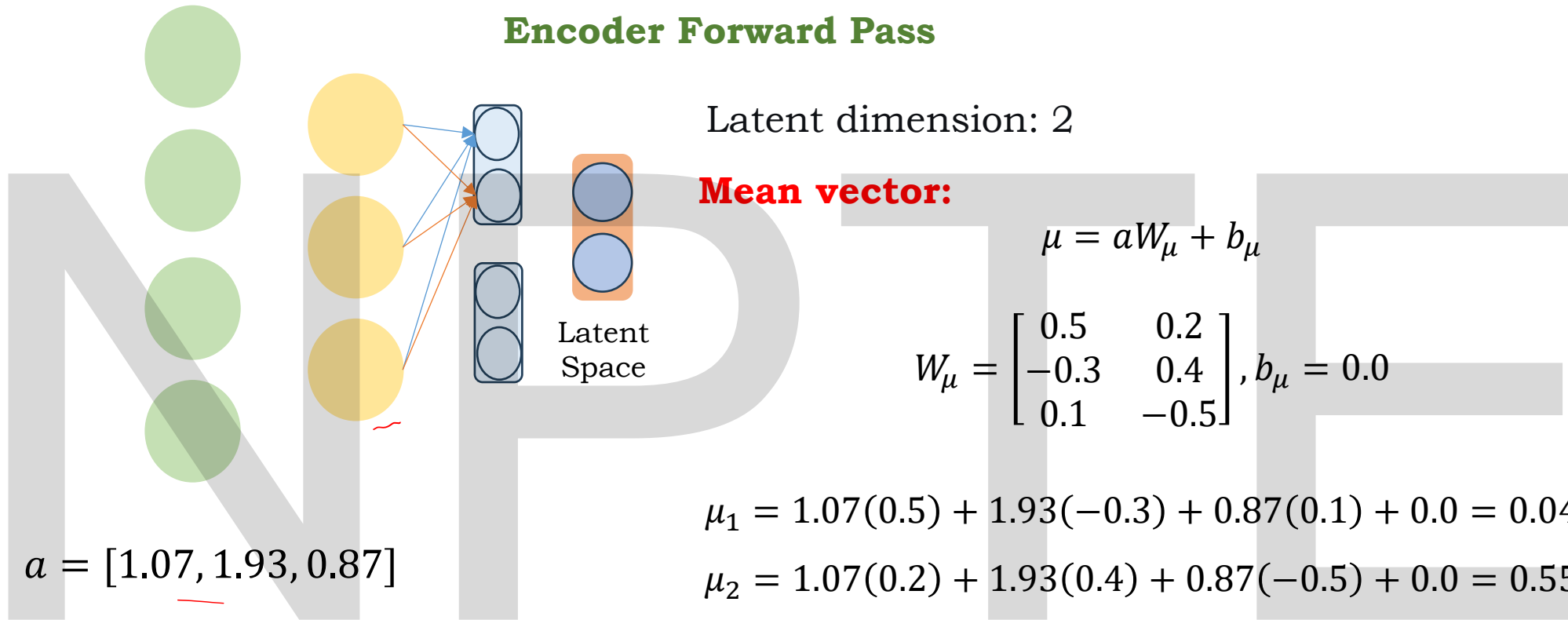
$$h_1 = 1.07, h_2 = 1.93, h_3 = 0.87$$

the ReLU output remains the same:

$$\underline{a = [1.07, 1.93, 0.87]} \checkmark$$

Variational Autoencoders (VAEs): Encoder

Encoder Forward Pass



No standard activation function is applied to the **mean (μ)** $\mu = [0.043, 0.551]$ layer of a VAE. The mean can be: Positive, negative, or zero
For example:

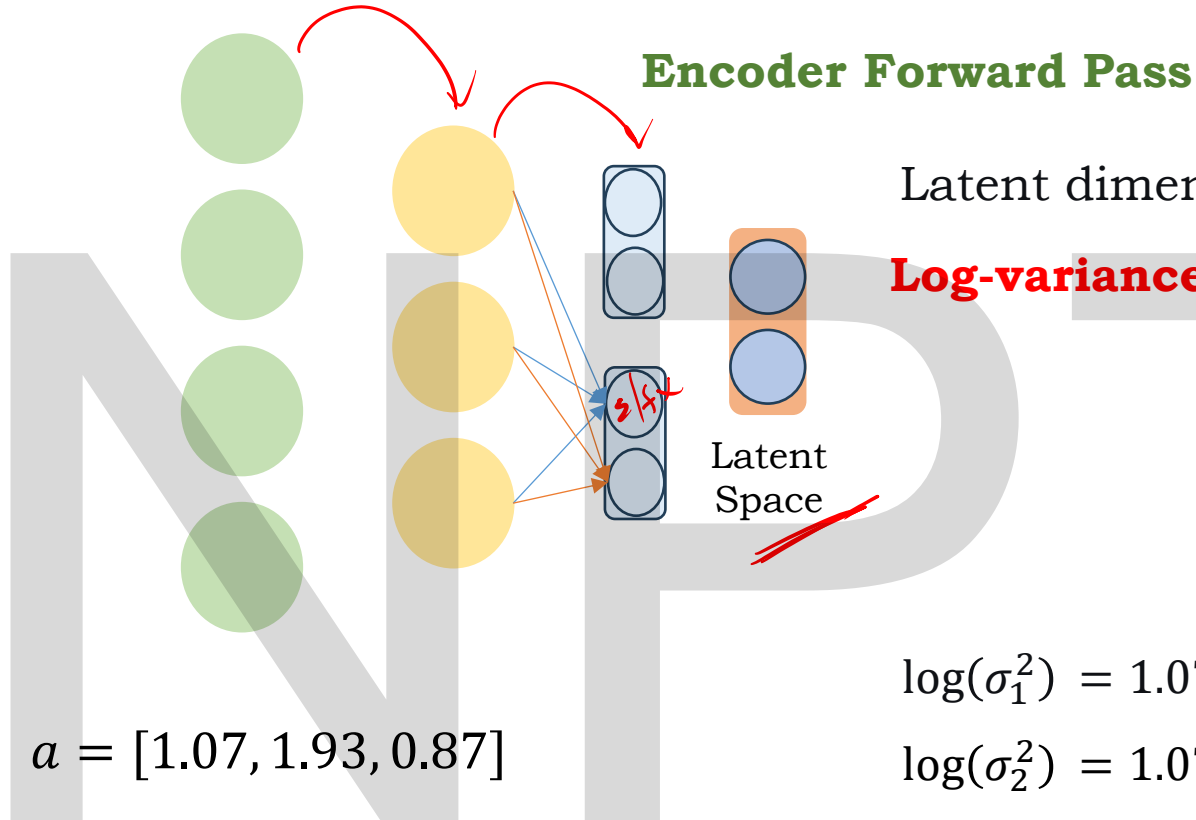
$$\mu = [-2.5, 0.8]$$

is perfectly valid.

If we apply ReLU:

$$\text{ReLU}(-2.5) = 0$$

Variational Autoencoders (VAEs): Encoder



$$\log(\sigma^2) = aW_{\sigma} + b_{\sigma}$$

$$W_{\sigma} = \begin{bmatrix} -0.2 & 0.3 \\ 0.4 & -0.1 \\ 0.2 & 0.2 \end{bmatrix}, b_{\sigma} = 0.0$$

$$\log(\sigma_1^2) = 1.07 \times (-0.2) + 1.93 \times 0.4 + 0.87 \times 0.2 + 0.0 = 0.732$$

$$\log(\sigma_2^2) = 1.07 \times 0.3 + 1.93 \times (-0.1) + 0.87 \times 0.2 + 0.0 = 0.302$$

$$\log(\sigma^2) = [0.732, 0.302]$$

Similarly, No standard activation function is applied to the **log variance** ($\log \sigma^2$) layer of a VAE.

Convert to Variance and Standard Deviation:

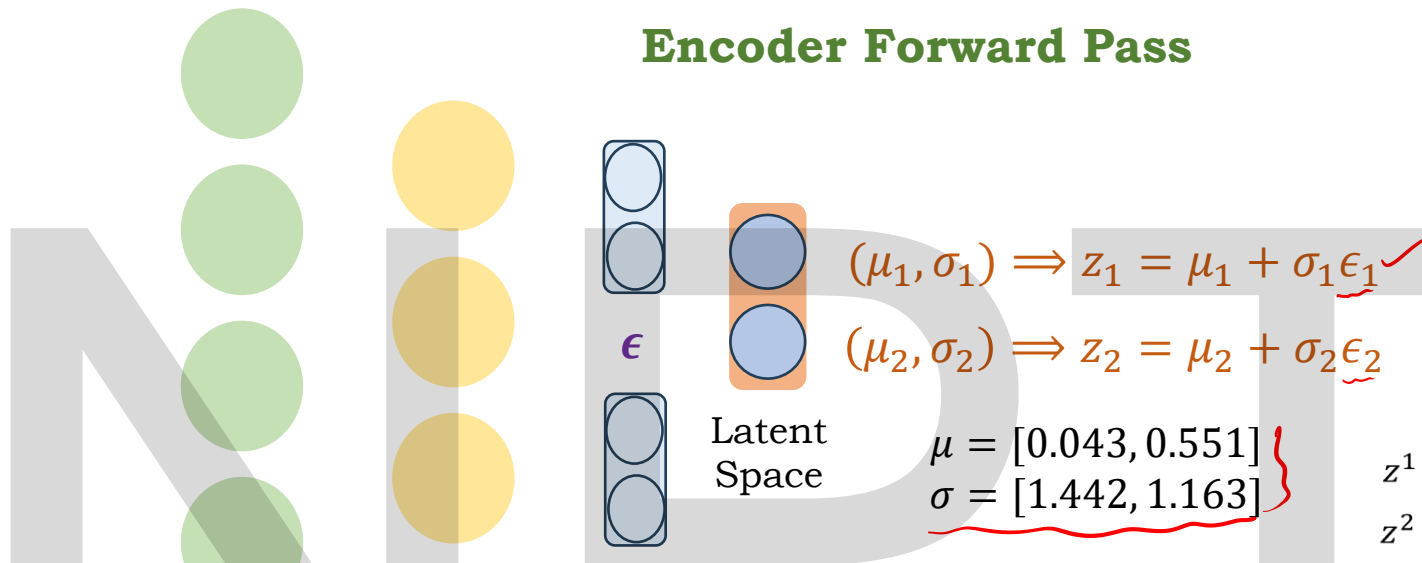
$$\sigma^2 = e^{\log(\sigma^2)} \quad \sigma = \sqrt{\sigma^2}$$

$$\sigma_1^2 = e^{0.732} = 2.079, \quad \sigma_2^2 = e^{0.302} = 1.353$$

$$\sigma_1 = 1.442, \quad \sigma_2 = 1.163$$

Variational Autoencoders (VAEs): Sampling from Latent space

Encoder Forward Pass



For the same input x , the encoder produces one distribution $q_\phi(z | x)$.

Different values of ϵ give different sampled latent vectors z from this same distribution.

	z_1	z_2
z^1	0.764	0.318
z^2	0.043	0.551
z^3	0.908	0.086
$\vdots$		
z^n		

Since:

$$\epsilon = \mathcal{N}(0,1)$$

Suppose:

$$\epsilon^{(1)} = [\epsilon_1, \epsilon_2] = [0.5, -0.2]$$

Then:

$$z_1^{(1)} = 0.043 + 1.442(0.5)$$

$$z_1^{(1)} = 0.043 + 0.721 = 0.764$$

$$z_2^{(1)} = 0.551 + 1.163(-0.2)$$

$$z_2^{(1)} = 0.551 - 0.233 = 0.318$$

Therefore:

$$z^{(1)} = [0.764, 0.318]$$

Suppose:

$$\epsilon^{(2)} = [\epsilon_1, \epsilon_2] = [0, 0]$$

Then:

$$z_1^{(2)} = 0.043 + 1.442(0)$$

$$z_1^{(2)} = 0.043 + 0 = 0.043$$

$$z_2^{(2)} = 0.551 + 1.163(0)$$

$$z_2^{(2)} = 0.551 + 0 = 0.551$$

Therefore:

$$z^{(2)} = [0.043, 0.551]$$

Suppose:

$$\epsilon^{(3)} = [\epsilon_1, \epsilon_2] = [0.6, -0.4]$$

Then:

$$z_1^{(3)} = 0.043 + 1.442(0.6)$$

$$z_1^{(3)} = 0.043 + 0.865 = 0.908$$

$$z_2^{(3)} = 0.551 + 1.163(-0.4)$$

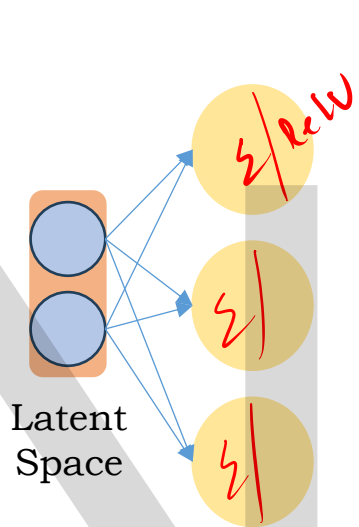
$$z_2^{(3)} = 0.551 - 0.465 = 0.086$$

Therefore:

$$z^{(3)} = [0.908, 0.086]$$

$z^{(1)}, z^{(2)}, z^{(3)}$ are sampled from $q_\phi(z | x)$

Variational Autoencoders (VAEs): Decoder



Latent Space

Hidden neuron h_1

$$h_1 = (0.764 \times 0.6) + (0.318 \times 0.3) + 0.1 \quad \checkmark$$

$$h_1 = 0.4584 + 0.0954 + 0.1 = 0.6538$$

$$a_1 = \text{ReLU}(h_1) \quad \checkmark$$

$$\underline{a_1} = \max(0, 0.6538) = 0.6538 \quad \checkmark$$

Hidden neuron h_2

$$h_2 = (0.764 \times -0.4) + (0.318 \times 0.1) + 0.1$$

$$h_2 = -0.3056 + 0.0318 + 0.1 = -0.1738$$

$$a_2 = \text{ReLU}(h_2)$$

$$\underline{a_2} = \max(0, -0.1738) = 0 \quad \checkmark$$

Hidden neuron h_3

$$h_3 = (0.764 \times \underline{0.2}) + (0.318 \times \underline{-0.5}) + 0.1 \quad \checkmark$$

$$h_3 = 0.1528 - 0.159 + 0.1 = \underline{0.0938}$$

$$a_3 = \text{ReLU}(h_3)$$

$$\underline{a_3} = \max(0, 0.0938) = \underline{0.0938}$$

The decoder architecture is:

$z(2 \text{ units}) \rightarrow h(3 \text{ units})$

You have sampled:

$$z = [0.764, 0.318] \quad \checkmark$$

Decoder hidden layer:

$$h = zW_2 + b_2$$

Given:

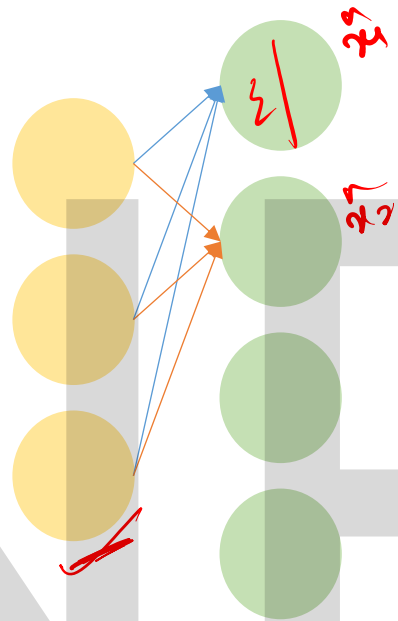
$$W_2 = \begin{bmatrix} \underline{0.6} & \underline{-0.4} & \underline{0.2} \\ \underline{0.3} & \underline{0.1} & \underline{-0.5} \end{bmatrix}$$

$$b_2 = 0.1$$

Since this is a decoder hidden layer, an activation function is applied after calculating h . Here it is ReLU.

$$\underline{\therefore a} = [\underline{0.6538}, 0, \underline{0.0938}]$$

Variational Autoencoders (VAEs): Decoder



Latent Space

The decoder architecture is:
 $h(3 \text{ units}) \rightarrow \hat{x}(4 \text{ units})$

Given:

$$W_3 = \begin{bmatrix} 0.2 & -0.1 & 0.3 & 0.4 \\ 0.1 & 0.2 & -0.2 & 0.1 \\ -0.3 & 0.4 & 0.1 & -0.2 \end{bmatrix}$$

$$b_3 = 0.0$$

$$\hat{x} = \mu_x = aW_3 + b_3$$

Linear activation function is used

Since this is the **decoder output layer**, What activation function to apply depends on the type of data.

For a continuous data, we use linear activation function:

$$\therefore \hat{x} = f(aW_3 + b_3) = aW_3 + b_3$$

If the input values are normalized between 0 and 1 or for the binary data, then sigmoid can be applied:

$$\hat{x} = \text{sigmoid}(aW_3 + b_3)$$

Output $\hat{x}_1$

$$\hat{x}_1 = (0.6538 \times 0.2) + (0 \times 0.1) + (0.0938 \times -0.3)$$

$$\hat{x}_1 = 0.13076 + 0 - 0.02814 = 0.10262$$

$$f(0.10262) = 0.10262$$

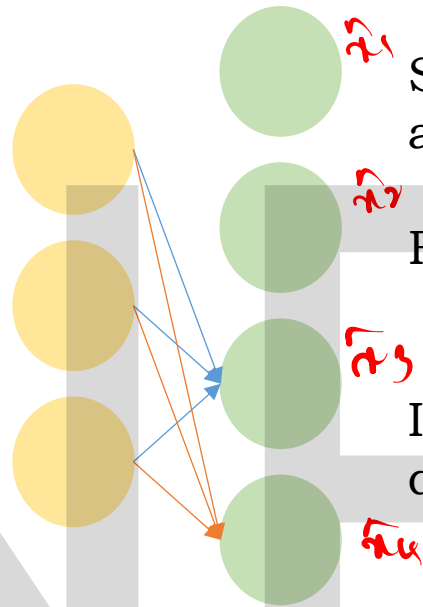
Output $\hat{x}_2$

$$\hat{x}_2 = (0.6538 \times -0.1) + (0 \times 0.2) + (0.0938 \times 0.4)$$

$$\hat{x}_2 = -0.06538 + 0 + 0.03752 = -0.02786$$

$$f(-0.02786) = -0.02786$$

Variational Autoencoders (VAEs): Decoder



Latent Space

The decoder architecture is:
 $h(3 \text{ units}) \rightarrow \hat{x}(4 \text{ units})$

Given:

$$W_3 = \begin{bmatrix} 0.2 & -0.1 & 0.3 & 0.4 \\ 0.1 & 0.2 & -0.2 & 0.1 \\ -0.3 & 0.4 & 0.1 & -0.2 \end{bmatrix}$$

$$b_3 = 0.0$$

$$\hat{x} = \mu_x = aW_3 + b_3$$

Linear activation function is used

Since this is the **decoder output layer**, What activation function to apply depends on the type of data.

For a continuous data, we use linear activation function:

$$\therefore \hat{x} = f(aW_3 + b_3) = aW_3 + b_3$$

If the input values are normalized between 0 and 1 or for the binary data, then sigmoid can be applied:

$$\hat{x} = \text{sigmoid}(aW_3 + b_3)$$

Output $\hat{x}_3$

$$\hat{x}_3 = (0.6538 \times 0.3) + (0 \times -0.2) + (0.0938 \times 0.1)$$

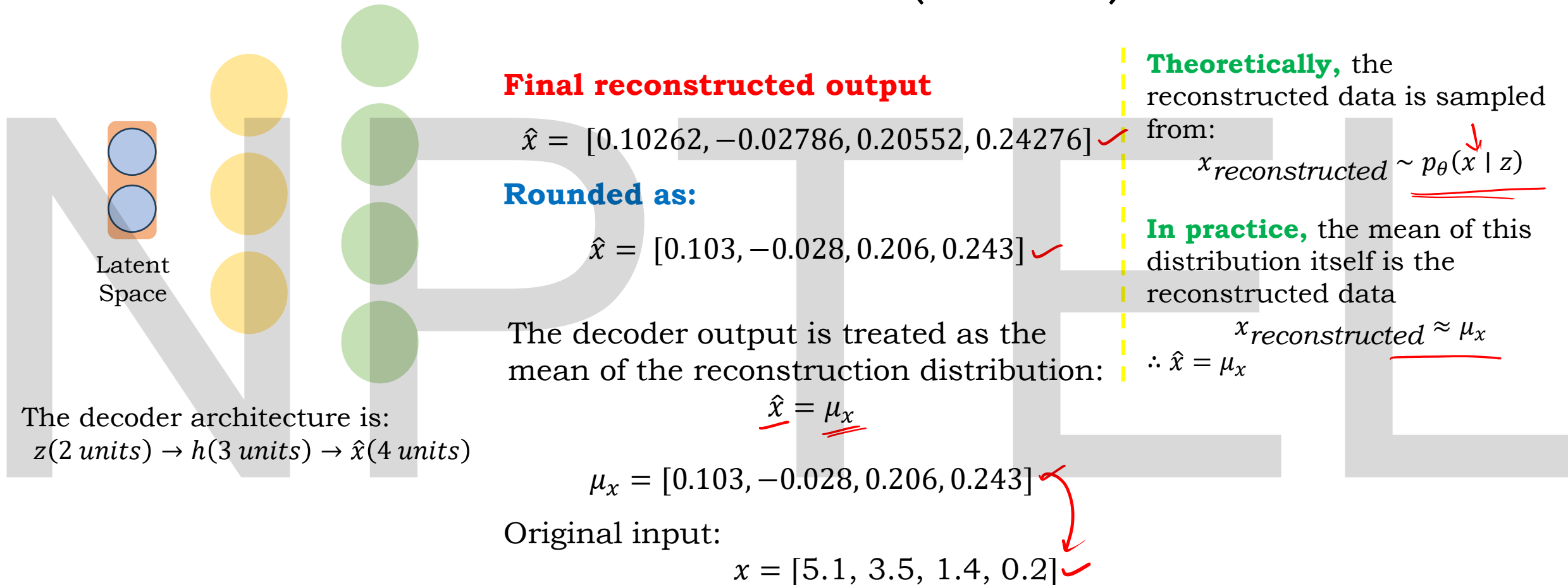
$$\hat{x}_3 = 0.19614 + 0 + 0.00938 = 0.20552$$

Output $\hat{x}_4$

$$\hat{x}_4 = (0.6538 \times 0.4) + (0 \times 0.1) + (0.0938 \times -0.2)$$

$$\hat{x}_4 = 0.26152 + 0 - 0.01876 = 0.24276$$

Variational Autoencoders (VAEs): Decoder



The reconstruction is poor because the weights and biases are randomly initialized. After training, the learnable parameters are optimized so that $\hat{x}$ becomes closer to x .

Variational Autoencoders (VAEs): Loss

Reconstruction Loss

Since the input is the continuous data, we compute:

Given:

$$\text{Reconstruction Loss} = \frac{1}{4} \sum_{i=1}^4 (x_i - \mu_{x,i})^2$$

$$x = 5.1, 3.5, 1.4, 0.2] \checkmark$$

$$\hat{x} = \mu_x = [0.103, -0.028, 0.206, 0.243] \checkmark$$

$$\begin{aligned} \text{Reconstruction Loss} &= \frac{1}{4} \left[(5.1 - 0.103)^2 + (3.5 - (-0.028))^2 + (1.4 - 0.206)^2 + (0.2 - 0.243)^2 \right] \\ &= \frac{1}{4} [24.9700 + 12.4468 + 1.4256 + 0.0018] \\ &= \frac{38.8443}{4} \end{aligned}$$

$$\text{Reconstruction Loss} = 9.7111$$

Variational Autoencoders (VAEs): Loss

KL Divergence Loss

The KL divergence is:

$$L_{KL} = KL(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, 1))$$

$$L_{KL} = \frac{1}{2} \sum_{j=1}^2 (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Compute:

$$\mu_1^2 = (0.043)^2 = 0.00185$$

$$\mu_2^2 = (0.551)^2 = 0.3036$$

$$KL = \frac{1}{2} [(0.00185 + 2.079 - 0.732 - 1) + (0.3036 + 1.353 - 0.302 - 1)]$$

$$= \frac{1}{2} [0.34885 + 0.3546] = \frac{1}{2} (0.70345) = 0.3517$$

Rounded:

$$KL \approx 0.352$$

Encoder Outputs

$$\mu = [0.043, 0.551]$$

$$\log \sigma^2 = [0.732, 0.302]$$

$$\sigma^2 = [2.079, 1.353]$$

$$\begin{aligned} \text{Total VAE Loss} &= \text{Reconstruction Loss} + \text{KL Divergence} \\ &= 9.7111 + 0.352 \\ &= \underline{10.0631} \end{aligned}$$

Next Session

- In week 4 (1st session): Hands-on implementation of Variational Autoencoders

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Thank you